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DETAIL SPECIFICATION

Cup, Water Canteen, W/Wire Handle, Corrosion-Resisting Steel

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers one type and size of corrosion-resisting steel canteen cup with a folding wire handle (see 6.1).

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they should meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether they are listed or not.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, DLA Troop Support Clothing and Textile, 700 Robbins Ave, Philadelphia, 19111-5096.

AMSC N/A FSC 8465

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SPECIFICATIONS

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-12332 - Welding, Resistance, Spot, Seam, And Projection; For Fabricating Assemblies Of Low-Carbon Steel

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Military Marking for Shipment and Storage

Copies of the above noted documents are available online at https://assist.dla.mil.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

NATICK LABORATORY ARMY RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-9-284 Cup, Canteen; Assembly and Details

Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQ Z1.4 - Sampling Procedures And Tables For Inspection By Attributes

Copies of this document are available online at https://asq.org.

AMERICAN WELDING SOCIETY

AWS D17.1/D17.1M - Fusion Welding for Aerospace Applications, Specification for

Copies of this document are available online at http://www.aws.org.

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ASTM International

ASTM A240/A240M	-	Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications, Standard Specification for
ASTM A967/A967M	-	Chemical Passivation Treatments for Stainless Steel Parts, Standard Specification for
ASTM E18	-	Rockwell Hardness of Metallic Materials, Standard Test Methods for

Copies of these documents are available online at https://www.astm.org.

SAE international

AMS QQ-S-763	-	Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings

Copies of this document are available online at https://www.sae.org.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Toxic chemicals, hazardous substances, and ozone-depleting chemicals. The use of toxic chemicals, hazardous substances, or ozone-depleting chemicals shall be avoided, if feasible. The desired performance requirements should be specified rather than the specific chemical or substance. If a toxic chemical, hazardous substance, or ozone-depleting chemical shall be specified, it shall be listed as a key word in section 6. The Environmental Protection Agency maintains an online list of toxic chemicals and hazardous substances at http://www.epa.gov/emergencies/tools.htm#lol that should be consulted.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials and components. The materials and components shall be as specified herein and on Drawing 2-9-284. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document.

3.4.1 Steel, corrosion-resisting.

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3.4.1.1 Sheet and strip. Corrosion-resisting steel sheet and strip for cup, (body and hinge), shall be class 302 or 304 of ASTM A240/A240M.

3.4.1.2 Wire. Corrosion-resisting steel wire shall conform to condition B, class 302 or 304, form round, finish cold of AMSQQS763.

3.5 Design and construction. The design and construction of the canteen cup shall conform to Drawing 2-9-284 and the requirements specified herein (see 6.4).

3.5.1 Body. The body of the cup shall be fabricated of the corrosion-resisting steel specified in 3.4.1.1 and shall be of seamless construction conforming to the dimensions specified on Drawing 2-9-284. The top edge shall be free of sharp edges, cracks, burrs, slivers, or rough surfaces. After forming, the vertical walls of the body shall have a minimum hardness of 46 on the 30N Rockwell scale when tested as specified in 4.5.5.

3.5.2 Handle. The handle shall be fabricated of corrosion-resisting steel specified in 3.4.1.2 and shall be welded in accordance with AWS-D17.1 or MIL-W-12332. The resistance "T" welds of the handle shall withstand a minimum total force of 235 pounds when tested as specified in 4.5.4. The handle shall conform to the design, dimension, and detail requirements of Drawing 2-9-284. The handle shall fit snugly against the cup within 1/8 inch when folded and the right handle shall rest and lock into the left handle when in use.

3.5.3 Hinge. The hinge shall be fabricated of corrosion-resisting steel specified in 3.4.1.1 and shall conform to the design, dimension, and details specified on Drawing 2-9-284. The hinge shall be attached by spot welds or projection welds in accordance with MIL-W-12332 and shall be located on the cup as shown on Drawing 2-9-284. The welds shall be free of splatter, burn holes, cracks or fractures, and discoloration and shall be tightly joined when tested as specified in 4.5.1, 4.5.2, or 4.5.3, as applicable. The hinge shall be a close fit to the contour of the body and there shall be no gaps between edge of hinge and body greater than 1/64 inch.

3.6 Finish. The canteen cup shall be given a uniform frosted lusterless finish approximating a commercial 2D sheet finish and passivated in accordance with AMSQQS763 or ASTM A967/A967M. There shall be no evidence of rust, staining, pitting, or spotting when tested as specified in 4.5.6.

3.7 Marking. The letters US, the year of manufacture, and the manufacturer's name or trademark shall be stamped on the hinge as shown on Drawing 2-9-284.

3.8 Workmanship. The finished canteen cup shall be clean and free of any dent or being bent out of shape, stain, ripple, fracture, die or draw mark. The handles and hinge shall also be free of any sharp edge, burr, or sliver. Indentation on the inside of cup due to welding of the hinge is acceptable providing the indentation is smooth and free from sharp edges, weld splatter, burr, crack, or burn through area, and does not exceed 0.005 inch in depth.

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4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the appropriate paragraph below.

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4.

4.4 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. In-process examination shall be made to establish conformance to the welding on wire handles requirements of Drawing 2-9-284. Whenever nonconformance is noted, correction shall be made to the applicable process and all items processed.

4.4.2.2 In-process spot or projection weld testing. The spot or projection weld sample peel test shall be conducted on two test specimens per welding machine rather than sample cups. The spot or projection welding testing shall be performed as specified in 4.5.1 at the start of each day's shifts production. Failure of test shall require correction of the welding process.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of cups. The sample unit shall be one cup. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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Table I
End Item visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not a uniform frosted, lusterless finish approximating a commercial 2D sheet finish		201
	Evidence of rust, staining, pitting, or spotting (slight water spotting, easily removed by washing shall not be considered a defect)	101	
Construction and Workmanship (applicable to all components and assemblies)	Seam, lap, laminations or crease, or metal ripple	102	
	Any component missing	103	
	Component dented		202
	Component fractured	104	
	Component bent out of shape		203
	Rim of cup contains sharp edge, burr, crack, sliver or rough spot	105	
	Evidence of die or draw mark		204
	Handle does not fold around cup within 1/8 inch as specified, or right handle does not nest into left handle	106	
	Hinge plate is not close fit against body, i.e. contains gap along edge greater than 1/64 inch (use feeler gage)		205
	Handle binds or requires force in opening or folding		206
	Handle or hinge contains sharp edge, burr, or sliver	107	
Welds	Contains crack or fracture	108	
	Not number specified	109	
	Not in location shown on drawing		207
	Contains burn hole, splatter, or discoloration	110	
	Evidence of poor fusion	111	
	Any missing weld	112	
	Weld on handle not neatly performed; e.g., contains rough or sharp projection or bead		208
	Indentation in weld area on inside of cup (Indentation on inside of the cup due to welding of the hinge is acceptable providing the indentation is smooth and free from sharp edges, weld splatter, burr, crack, burn through area, and does not exceed 0.005 inch in depth)		209
Marking	Missing, illegible or incomplete	113	

4.4.4 End item dimensional examination. The end items shall be examined for the dimensions annotated with an asterisk on Drawing 2-9-284. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of cups. The sample unit shall

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be one cup. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items shall be tested as specified in 4.5.2 or 4.5.3, 4.5.4, 4.5.5, and 4.5.6. The lot size shall be expressed in units of cups. The sample unit shall be one cup. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5 4.5 Methods of inspection.

4.5.1 Spot or projection weld sample peel test. Each spot, or projection weld test specimen shall be of the type and thickness of material used in the cup body and hinge. Spot or projection weld each section as shown on Drawing 2-9-284 and then place in a vise or jig. Bend the parent metal piece and peel the hinge piece with pliers or another suitable tool. Inability of any spot weld to pull a 5/32-inch (minimum) hole in either piece of material shall constitute failure of this test. Failures of the projection welds or lack of fusion under the projection weld shall constitute failure of the test. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion.

4.5.2 Spot weld cup peel test. Place the cup in a jig or vise and peel the hinge from the cup. Inability of each weld to leave a 5/32-inch (minimum) hole in either the cup or hinge shall constitute a defect.

4.5.3 Projection weld cup peel test. Separate the weldment of the cup and hinge by driving a chisel between welded components in an unwelded area or by peeling the hinge back against the weld until failure occurs. Failure outside the periphery of the weld area shall be considered evidence that the welds are satisfactory. Lack of fusion under the projection weld shall constitute a defect. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion.

4.5.4 Test for resistance "T" welds of the handle. Place the cup in a jig or vise and exert a force against both "T" welds of the handle. Inability of any weld to withstand the minimum force specified in 3.3.2 shall constitute a defect.

4.5.5 Rockwell hardness test. Test the Rockwell hardness of the vertical walls of the body of the cup in accordance with ASTM E18. Take four Rockwell hardness readings on the vertical walls each 90 degrees apart around the cup and approximately half-way between the top and the bottom of the cup. Any sample cup having one or more readings less than 46 on the Rockwell 30N scale as specified in 3.3.1 shall be classified as a defect.

4.5.6 Finish test. Expose the complete assembled cup to 95-100 percent relative humidity at 72° ± 2°F for not less than 16 hours to determine compliance with rust, staining, pitting, and spotting requirements of 3.4. Inability of the cup to conform to requirements of 3.4 shall constitute a defect.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-

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house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The corrosion-resisting steel cups are for use with canteen, water, plastic, MIL-C-43103. The cup is used for drinking hot and cold liquids, soups, and stews in the field.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required (see 3.1, 4.3, and 6.3).
d. Packaging requirements (see 5.1).

6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Hydroforming. Hydroforming has been found to produce acceptable cups.

6.5 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.6 Subject term (key word) listing.

Chemical protection
Clothing, special purpose
Parachutist's helmet

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensive nature of the changes.

Custodians:
Army – GL
Navy – NU
Air Force – 11

Preparing Activity:
DLA – CT
Project 8465-2023-008

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Review activities:
Army – MD
Navy – CG, MC

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